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Mitigating food waste and household waste management: The potential for redistributing surplus food in the policy communication of Medan City government

Feni Khairifa^{1*}, Syukur Kholil², Abdi Mubarak Syam² and Naqil Sayyaf Al-Mujahid²

¹ Faculty of Social and Political Sciences, Universitas Sumatera Utara, Medan 20155, Indonesia

² Universitas Islam Negeri Sumatera Utara, Medan, Indonesia

*E-mail: fenikhairifa@usu.ac.id

Abstract. This article aims to examine strategies for mitigating food waste and managing household waste in Medan City through the potential redistribution of surplus food within the context of public policy communication. The study adopts a qualitative approach using a case study method, focusing on understanding the phenomenon of surplus food redistribution from the perspective of various stakeholders, including local government, NGOs, food businesses, and local communities. Primary data sources were collected through in-depth interviews and participatory observations, while secondary data were obtained from official documents and activity reports. The research findings indicate that waste reduction policies implemented through Medan Mayor Regulation No. 26 of 2019 have successfully increased public awareness and participation in waste management. This policy includes education and public campaigns, training, and cooperation with local institutions and businesses, as well as incentive programs that encourage active community involvement. Coordination and collaboration between government sectors and the private sector also strengthen the implementation of this policy, ensuring well-coordinated waste management efforts. Surplus food redistribution in Medan has shown effectiveness in reducing food waste and supporting local food security. Initiatives such as the Surplus application and the Food Cycle Indonesia program leverage technology and community cooperation to distribute consumable food to needy groups. This approach not only reduces the volume of waste but also supports the Sustainable Development Goals (SDGs), including Zero Hunger, Responsible Consumption and Production, and Climate Action. A clear and effective policy communication strategy, supported by cross-sector collaboration, can significantly contribute to reducing food waste and sustainably managing household waste in Medan City. This research provides valuable insights for developing similar policies in other regions.

1. Introduction

Childhood in Indonesia is often marked by the encouragement to finish food. This tradition aims to teach the importance of not wasting food, yet ironically, food waste practices are still prevalent. According to data from the Ministry of Environment and Forestry (KLHK) in 2023, Indonesia produces 13.37 million tons of waste annually, with 38.9% coming from households, of which



40.96% is food waste. This issue indicates a lack of communicative education in government policy dissemination.

A study by the National Development Planning Agency (Bappenas), in collaboration with Waste4Change and the World Research Institute in 2021, revealed that food waste in Indonesia reached 115-184 kg per capita per year from 2000-2019. This amount indicates a significant potential for utilizing food waste to meet the nutritional needs of 61-125 million Indonesians annually [1].

The economic impact of food waste is significant, estimated at 213-551 trillion rupiahs per year, and contributes approximately 7.29% of annual greenhouse gas emissions. Without strategic changes, food waste is projected to increase to 344 kg per capita by 2045, an almost 200% increase from the last two decades. Overconsumption behavior has become a damaging norm, indicating indifference to hunger issues and environmental overload [2]. Too often, people tend to buy more than they need, triggering a cycle of excessive consumption and disposal. This not only creates pressure on waste management systems but also wastes resources that could be used more effectively elsewhere. Policies that encourage purchases based on need, rather than desire, need to be strengthened to address this issue [3,4].

Initiatives like the Surplus application by PT Ekonomi Sirkuler Indonesia have started leveraging technology to tackle this problem. This app enables the redistribution of consumable food, supporting Sustainable Development Goals (SDGs) related to Zero Hunger, Responsible Consumption and Production, and Climate Action. The Yayasan Daur Pangan Nusantara has also taken similar steps through the FoodCycle Indonesia program, which collects surplus food from various businesses and distributes it to vulnerable communities, helping to reduce the amount of wasted food [5,6].

To address this issue, various regulations have been implemented at the local level. The North Sumatra Governor's Regulation (PERGUB) No. 3 of 2020, Regional Regulation (PERDA) No. 11 of 2023, and Medan Mayor Regulation (PERWALI) No. 26 of 2019, all regulate strategies for waste reduction and management, including food waste, at the regional and city levels (Database Peraturan | JDIH BPK) (Sumutprov) (Database Peraturan | JDIH BPK). All parties need to collaborate to support these strategies, from individuals, businesses, to the government. With a comprehensive approach and active participation from all elements of society, addressing food waste issues in Indonesia, particularly in Medan City, can become more effective [7].

The Medan City Government has taken policy communication initiatives focused on increasing public awareness and participation in household waste management and surplus food redistribution [8]. This strategy includes elements of education and cross-sector collaboration to implement programs that can significantly reduce food waste. In this context, policy communication is not merely about conveying information but also about building understanding and support from the community for the policies implemented [9].

The urgency of this research is reflected in the increasing volume of food waste that not only negatively impacts the environment through increased greenhouse gas emissions but also causes significant economic losses. This study is important to identify effective solutions in managing and reducing food waste in Medan City, while also exploring the potential of surplus food redistribution as a concrete step to support environmental sustainability and food security. Thus, the results of this research are expected to provide recommendations to the government and stakeholders on food waste reduction strategies, which will directly help achieve several Sustainable Development Goals (SDGs), including Zero Hunger, Responsible Consumption and

Production, and Climate Action. This is not only important for Medan or Indonesia but also provides insights that can be applied in other regions with similar conditions.

2. Methods

This research adopts a qualitative approach using a case study method to explore the topic holistically and contextually. Within a constructivist framework, this study focuses on understanding the phenomenon of surplus food redistribution in Medan City from the perspective of various stakeholders. This is done to construct a broader understanding of the social, economic, and environmental dynamics and implications of this practice [10].

Primary data sources were collected through in-depth interviews and participatory observations with parties directly involved in the surplus food redistribution process, including local government, NGOs, food businesses, and local communities. Secondary data were obtained from official documents, activity reports, and scientific publications relevant to the research topic. The use of diverse data sources is intended to obtain a comprehensive and in-depth perspective on the issue being studied.

This data collection technique was followed by data analysis using the Miles, Huberman, and Saldana model, modified with a data condensation process to facilitate the identification of emerging themes and patterns. This data condensation involves simplifying, merging, and sorting raw data to produce more concise and focused descriptions. This analysis process helps organize the data so it can be easily analyzed and interpreted [11].

The validity of the data in this study is strengthened through source triangulation techniques, where information obtained from different data sources is confirmed and compared to ensure the reliability and validity of the findings. This triangulation not only enhances the credibility of the research results but also enriches the analysis by allowing various perspectives and dimensions of the same issue to be examined further [12].

3. Result and discussion

The results of the interviews indicate that the Medan City Government has taken strategic steps to implement policies supporting surplus food redistribution. These policies aim not only to reduce the volume of waste produced but also to increase public awareness and participation in more responsible consumption practices. The city government recognizes that these interventions are essential to strengthening environmental sustainability and supporting local food security [13].

Surplus food redistribution in Medan involves collaboration between various sectors, including local government, food businesses, non-profit organizations, and communities [14]. This initiative begins with identifying sources of surplus food, including restaurants, markets, and supermarkets, to then be distributed to groups in need. This approach not only reduces food waste but also serves as a practical solution to combat urban hunger [9,15].

To address this issue, the Medan City Government has issued Mayor Regulation (Perwal) No. 26 of 2019 regarding regional policies and strategies for household waste management and similar waste in Medan City [16]. This regulation includes several communication policy aspects that can be analyzed, namely:

Table 1. Medan City government communication policies to address food waste and household waste issues.

Policy Aspect	Communication Strategy	Description	Expected Impact
Waste Reduction	Education and Public Campaigns	The Medan city government uses social media, workshops, and public campaigns to educate the public about the importance of reducing waste.	Increase public awareness of the importance of waste reduction, leading to a direct decrease in waste generation.
Waste Management	Training and Cooperation	The government trains residents and collaborates with local institutions and businesses to implement effective waste management practices, such as sorting and composting.	Improve community waste management skills and integrate businesses into waste handling efforts, enhancing waste processing and recycling.
Community Engagement	Incentive Programs	Introducing incentives for individuals and groups active in waste reduction and management activities, such as tax discounts or public service benefits.	Encourage active community participation in waste reduction and management activities, strengthening community-based waste management.
Inter-Sector Coordination	Multi-Stakeholder Forum	Organizing forums involving all stakeholders, including government, businesses, and NGOs, to formulate and synchronize waste management efforts.	Ensure well-coordinated and integrated waste management efforts, strengthening overall policy effectiveness.

In an effort to tackle waste management issues in Medan City, Mayor Regulation No. 26 of 2019 has implemented an integral and multifaceted communication strategy. Intensive education and public campaigns are carried out through various media, including social media and workshops, to raise public awareness about the importance of reducing waste. This initiative aims not only to broaden public understanding of effective waste management but also to stimulate sustainable behavioral changes related to consumption and waste disposal [17,18].

Furthermore, the city government has introduced training and cooperation with local institutions and businesses as part of its waste management strategy [19]. Through this approach, residents are trained in effective waste management techniques, such as sorting and composting, while businesses are integrated into these efforts to create more holistic and integrated solutions [20]. This strategy not only enhances community skills in waste management but also ensures that these practices become part of daily operations on a community and business scale [21,22].

The implementation of incentive programs has also been a vital instrument in this communication strategy, where individuals and groups active in waste reduction and management activities are rewarded, such as with tax discounts or enhanced access to public services. This initiative aims to encourage proactive community participation in activities supporting waste management policies, transforming public response from passive to active actions supporting environmental sustainability.

Additionally, inter-sector coordination has been strengthened through multi-stakeholder forums involving government, businesses, and civil society organizations. These forums aim to formulate and synchronize waste management efforts, ensuring that all parties have a common understanding and strong commitment to the established goals. This, in turn, strengthens

cooperation and ensures that waste management initiatives are effectively coordinated and integrated at all levels.

These comprehensive and structured efforts reflect the Medan City Government's commitment to implementing sustainable and effective waste management principles. Through clear and effective communication policies, Mayor Regulation No. 26 of 2019 aims not only to improve physical infrastructure but also to build community capacity and strengthen inter-sector cooperation, which are all key to achieving sustainable and effective waste management. The analysis of Medan City Government's communication policies is highly relevant in shaping competent regulations, given the burgeoning household waste and food waste problem in Indonesia. The policy points relevant to surplus food distribution are as follows:

Table 2. Food waste management and surplus food redistribution.

Policy Aspect	Description	Relevance to Surplus Food Redistribution
Waste Reduction	This regulation outlines strategies and targets to reduce household and similar waste through approaches including waste limitation and reuse.	This highlights the potential to include surplus food as part of the waste reduction strategy, ensuring that consumable food is redistributed rather than wasted.
Waste Management	Establishes measures for waste handling including sorting, collection, transportation, processing, and final disposal.	Provides a framework to process surplus food sustainably, such as by composting or other low-impact uses.
Community Engagement	The policy mandates strengthening community involvement through education, which can include information on reducing food waste and supporting surplus food programs.	Can encourage active community participation in food redistribution initiatives, increasing awareness and participation in food waste reduction programs.
Coordination and Collaboration	Encourages coordination and cooperation between various levels of government and sectors, including businesses, in waste reduction and management.	This collaboration can support surplus food redistribution initiatives, integrating efforts from various sectors for greater effectiveness.

Mayor Regulation No. 26 of 2019 concerning household and similar waste management reflects a strategic approach to addressing food waste and general waste issues. This policy is important not only in the context of waste management but also in the context of public policy communication, which plays a crucial role in implementing and socializing sustainability principles to the public [23,24].

Waste reduction policies through waste limitation and reuse demonstrate the government's efforts to encourage the public to reduce waste at its source. This step is relevant to surplus food redistribution, which targets the reuse of still-consumable food that would otherwise be discarded. This not only reduces the burden on landfills but also supports food security and reduces economic burdens on the community [25,26].

Furthermore, the waste management strategy, which includes sorting, collection, transportation, processing, and final disposal, provides a clear operational framework [27,28]. In the context of surplus food, this policy can be directed to support initiatives such as composting or other uses for organic waste. This step is also vital in policy communication, where the government must actively explain and demonstrate these processes and their benefits to the public.

Community involvement is also a crucial component in the success of waste management policies. Through effective education, the community is encouraged not only to understand but also to actively participate in waste reduction and management efforts [29,30]. Effective public education can increase awareness and change behavior, directly supporting the success of surplus food redistribution and waste management initiatives.

Moreover, coordination and collaboration between government sectors and between the government and private sectors underscore the importance of synergy in addressing waste issues. This policy emphasizes the need for cross-sector cooperation to achieve effective waste reduction and management targets [31]. In the context of policy communication, clarity of roles and responsibilities of each party is vital to convey so that all involved parties can work harmoniously and efficiently.

Overall, this policy demonstrates the Medan City Government's commitment to applying sustainable and effective waste management principles. Through clear and structured communication policies, the city government aims to convey the importance of this policy not only for the environment but also for the social and economic welfare of the community. This approach is expected to increase public participation and strengthen a culture of sustainability in Medan.

The results of implementing this policy show significant improvements in household waste management in Medan City. By integrating circular economy principles in waste management, the Medan City Government has successfully created a more efficient system supporting food waste reduction efforts while enhancing the quality of life for the community. This step serves as an example of how public policy can play a role in changing public behavior and supporting sustainable environmental initiatives.

4. Conclusion

Research on this topic highlights several key conclusions that contribute to environmental management and social policy in Medan City. Firstly, the waste reduction policy implemented by the city government not only demonstrates a commitment to environmental sustainability but also reflects concrete efforts to promote responsible consumption. Through education and public campaigns, public awareness about the importance of reducing food waste has increased, directly contributing to the reduction of waste generation. Secondly, this policy has successfully leveraged collaboration between government and non-government sectors, including businesses and local communities, to implement effective household waste management and surplus food redistribution programs. Initiatives such as the Surplus app and Food Cycle Indonesia program exemplify how technology and community cooperation can be utilized to support Sustainable Development Goals (SDGs), such as Zero Hunger, Responsible Consumption and Production, and Climate Action. This approach is not only relevant for Medan City but can also be adapted by other regions facing similar issues related to waste management and food waste.

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